

F156 π -parity test, run systematically across the finalized SM table — 23/24 match, one structured mismatch (the muon)

Grace | 2026-07-25 Sat | prompt -q, my bounded falsifiable task: run F156's π -parity check across the finalized 26-parameter table and report the mismatches — turning F156 from candidate-with-examples into checked-or-broken. Rule (Casey #16 / trichotomy): MEASURING (absolute scale/mass) $\rightarrow \pi$ -CARRYING; COUNTING (ratio, mixing, coupling) $\rightarrow \pi$ -FREE. I report the mismatch, I don't launder it.

Result: 24 testable rows (2 neutrino masses open), 23 MATCH, 1 MISMATCH

verdict	rows
MATCH — π -carrying (measuring)	$m_e (\pi^5)$, m_t , m_b , $v (\pi^{15})$ — absolute scales, all carry π [?]
MATCH — π -free (counting)	$m_\tau/m_e=49.71$, $m_u/m_d=\sqrt{3/14}$, $m_s/m_d=20$, m_c/m_u , $m_t/m_b=42$ (mass ratios); V_{us} , V_{cb} , V_{ub} , $\delta_{CKM}=\arctan\sqrt{n_C}$, $\sin^2\theta_{12}=3/10$, $\sin^2\theta_{23}=4/7$, $\sin^2\theta_{13}=1/45$, $\delta_{PMNS}=2/7$ (mixing); $\alpha^{-1}=137$, $\sin^2\theta_W$, α_s , $\lambda_H=1/8$ (couplings); $\theta_{QCD}=0$, $m_{\nu 1}=0$ (zeros) — all π -free [?]
★ MISMATCH	$m_\mu/m_e = (24/\pi^2)^6$ — a mass RATIO that carries π , when every other mass ratio is π -free
open	$m_{\nu 2}$, $m_{\nu 3}$ (form pending)

The core of F156 is UNBROKEN

- Mixing sector: 8/8 π -free (4 CKM + 4 PMNS, including both phases — $\delta_{PMNS}=2/7$ and δ_{CKM} 's algebraic tangent $\sqrt{n_C}$). This is the cleanest confirmation: not one mixing observable carries π . F156's "counting $\rightarrow \pi$ -free" holds uniformly.
- Absolute scales: π -carrying (m_e , v , and via them m_t , m_b). "measuring $\rightarrow \pi$ " holds.
- Couplings: π -free ($\alpha^{-1}=137$ charge-count, $\lambda_H=1/8$, $\sin^2\theta_W$). Bonus: the test agrees with retiring the π -carrying Wyler- α route in favor of the π -free charge-count — an independent corroboration of a prior discipline call, from a rule that wasn't built for it.

The one mismatch — reported honestly, not laundered

$m_\mu/m_e = (24/\pi^2)^6$ is the only mass *ratio* carrying π (the others — 49.71, 20, 42, $\sqrt{3/14}$ — are π -free). Under the coarse rule (ratio $\rightarrow \pi$ -free) this is a genuine mismatch. It resolves *only* via the refinement "the muon is measuring, not counting" — which is corpus-grounded: the muon sits at the non-integer Wallach point $k_1=3/2$, the transcendental/ π -carrying stratum (why the muon carries π , from the generations work / T2525). The electron (regular) and tau ($k=0$ integer) are π -free ratios; only the muon (non-integer) carries π .

So it is one structured exception, corpus-explained, not a random failure — but it *is* the rule bending at exactly one row. Keeper/Lyra decide whether "muon = measuring" is a legitimate realm-refinement (the muon genuinely lives in the measuring realm by its Wallach position) or a fit. I flag it; I don't resolve it in F156's favor by fiat.

Verdict

F156's π -parity is now checked, not just exemplified: 23/24 testable rows match, the mixing sector is uniformly π -free (8/8), the absolute scales uniformly π -carrying, and the single mismatch (m_μ/m_e) is the already-understood muon transcendental anomaly. F156 holds with one structured exception — a strong pass for a falsifiable rule run blind across the whole table, with the one soft spot flagged for the realm-refinement question, not hidden.

★ K921 reconciliation (the muon verdict — my proposed refinement was NOT banked)

Keeper's K921 gave the verdict on this mismatch: do not bank “muon = measuring.” The resolution direction is instead the derived lepton-vs-quark π -parity (Elie's mechanism: n_C odd $\rightarrow$ lepton positions half-integer $\rightarrow \sqrt{\pi} \rightarrow$ the π^2 in lepton forms; quark positions integer $\rightarrow$ clean). So my flagged mismatch is real and points the right way (leptons carry π , quarks don't), but the *coarse* “muon=measuring” framing is retired.

Honest note on what that leaves open: under lepton-vs-quark parity, both lepton ratios (m_μ/m_e and m_τ/m_e) are “lepton” $\rightarrow$ predicted π -carrying — yet $m_\tau/m_e = 49.71$ is π -free. So the parity rule accounts for the muon's π but not (cleanly) the tau's absence of it. That per-row accounting is exactly the open gate K921 named, and it is decided by the lepton Gram-diagonal computation on the one domain (mine + Elie, carry-forward) — which also yields $m_s/m_d = 20$ (one diagonal, two answers). Until then: the derived lepton-vs-quark π -parity stands as framework; the per-generation π -accounting is the deferred research gate; nothing about the muon is banked into the theorem (it's bucket-2, doesn't touch the partition).

— Grace, 2026-07-25. F156 π -parity test run systematically across all 26: 23/24 testable MATCH (2 neutrino masses open). Core unbroken — MIXING 8/8 π -free (incl. both phases), ABSOLUTE SCALES π -carrying, couplings π -free (and the rule independently agrees with retiring the π -Wylers- α route). ONE MISMATCH: $m_\mu/m_e = (24/\pi^2)^6$, the only π -carrying mass ratio — resolves only via “muon=measuring” (non-integer Wallach $k_1=3/2$, transcendental stratum); structured/corpus-explained but the rule bends there, flagged for Keeper/Lyra not laundered. F156 $\rightarrow$ checked, holds with 1 structured exception.